

Patrick M. TIMONS

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EDUCATION

Artificial Intelligence And Decision Making — *Bachelor of Science*

Massachusetts Institute of Technology

AUG 2021 - MAY 2025

GPA: 4.7/5.0

- *Software Engineering*: Software Performance Engineering, Algorithms II, Algorithms I, Computer Programming, Computation Structures, Low-Level Programming, Mathematics for Comp Sci
- *Mathematics and Machine Learning*: Inference and Information (G), Machine Learning (G), Machine Learning (UG), Deep Learning (G), Deep Learning (UG), Probability, Statistics, Linear Algebra and Optimization, Representation, Inference, and Reasoning in AI, Quantitative NLP, Computer Vision
- *Economics and Social Science*: Networks, Microeconomics, Macroeconomics, Information Policy
- *Other Science*: Linguistics, Chemistry, Biology

WORK EXPERIENCE

Amazon Web Services — *SDE Intern*

JUN 2024 - AUG 2024

- Designed and implemented an end-to-end data pipeline to process real-time billing data of 550,000 enterprise AWS customers
- Built an ML-powered forecasting service for AWS cost-savings data analysis, served through an query-based API and a front-end dashboard
- Used AWS, Athena, IaC, CDK, SageMaker, S3, API Gateway, Data Catalog, Java, TypeScript, Python

Stealth Start-up — *Co-Founder, CTO*

SEP 2024 - MAR 2025

- MIT \$100K Finalist: Recognized among top MIT startups for innovation and market potential
- Winner of Harvard Law Entrepreneurship Competition: Gained access to mentorship from 8 Harvard Law School attorneys
- MIT Sandbox Grant Recipient: Awarded \$10,000 (\$5000 cash and \$5000 AWS Credits) for product development and scaling

Laboratory for Information and Decision Systems — *Research Assistant*

AUG 2022 - FEB 2023

- Spearheaded data collection initiative for fuel emission modeling project
- Generated simulated drive cycle data through use of MOVES
- Used Python, SQL and pandas to deposit simulated data in a MySQL database for downstream analysis

TECHNICAL PROJECTS

Improving Deep Learning Based Molecular Fingerprints Through Informed Resampling —

OCT 2023 - DEC 2023

- Worked in small team to research improvements to pretraining methods for transformer-based molecular encoders
- Created custom RoBERTa model through the use of transformers and bert-loves-chemistry (ChemBERTa) software packages with 9.1 percent improvement in Spearman’s rank correlation coefficient on downstream tasks compared to the base model
- Coauthored 5-page ACL-submission-style write-up about project and results

Recovering Latent Variables with Variational Autoencoders despite Training Bias —

NOV 2023 - DEC 2023

- Researched how beta-regularization robustifies VAEs to training bias when attempting to recover latent variables
- Trained models with PyTorch Lightning and used scientific computing libraries to generate training data and visualize results

Attentional Search —

JUN 2023 - JUL 2023

- Created search engine that uses attention mechanisms in transformer-based encoders to visualize search results
- Designed and implemented pipeline to retrieve and visualize data and created web interface with Dash

EXTRACURRICULAR ACTIVITIES AND HONORS

MIT Men’s Lacrosse — *Team Member*

AUG 2021 - MAY 2025

- Awarded USILA Academic All-American for the 2025 Season
- Received Ben Martin Award for “senior lacrosse player who has continuously exhibited the qualities of dedication, determination, and spirit”
- Voted Most Improved for 2023 season, NEWMAC All-Academic Award (2023, 2024, 2025), 2022 NEWMAC Champion

Global Teaching Labs — *Applied Math Teacher*

JAN 2024 - JAN 2024

- Teaching selected topics in Operations Research with a focus on Bayesian Inference to high school students in Cremona, Italy

SKILLS

Foundational Data Structures & Algorithms, Programming (Python, Java, TypeScript, C++), Full-Stack

Design Patterns and Programming Paradigms Model-first development, Dependency Injection, Object-Oriented-Programming, Procedural Programming, Concurrency, Multithreading

Dev Ops Source Control, Build Tools, Dependency Management, CI/CD

Databases & Batch Computing SQL, Apache Spark, AWS Data Catalog, Slurm Workload Manager

Libraries and Ecosystems React, AWS, SageMaker, PyTorch, Pytorch Lightning, Numpy, Pandas, SciKit Learn, Transformers, Transformers Reinforcement Library, Matplotlib, Plotly, Dash

Soft Skills Technical Due Diligence, Technical Writing